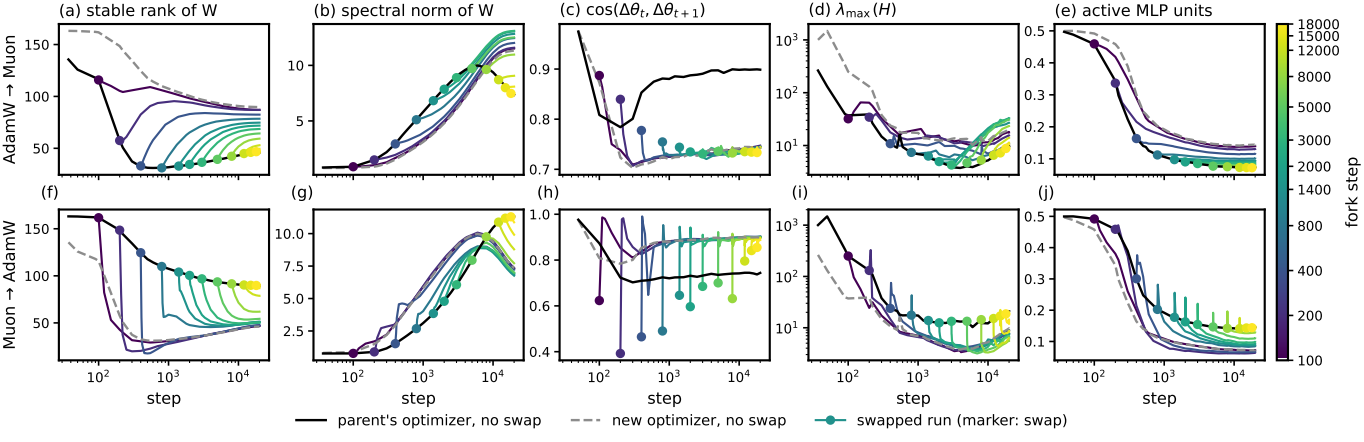

(a) stable rank of W
(b) spectral norm of W
(c) cos(Δθ_t, Δθ_{t+1})
(d) λ_max(H)
(e) active MLP units
(f)
(g)
(h)
(i)
(j)
AdamW → Muon
Muon → AdamW
step
fork step
18000
15000
12000
8000
5000
3000
2000
1400
800
400
200
100
parent's optimizer, no swap
new optimizer, no swap
swapped run (marker: swap)